CHE-225 Introduction to Chemical Engineering Analysis

Spring 2023

PROBLEM SET 8
(Due Mar 31, 2023 @ 8:30 AM in Gradescope)

This problem set is to be completed **individually** on a **hardcopy** and submitted via Gradescope. Make sure you start a blank workbook on your own computer. Do not forget to press the submit button in Moodle.

Problem 1 (**35 points total**)

Should you go to Japan one day, you have to spend some time in a typical Japanese bar and order sake (酒) (https://en.wikipedia.org/wiki/Sake), a traditional Japanese rice wine (the "*de jure*" legal drinking age in Japan is 20, so you may go there now ☺). In a classy Japanese bar, a server will bring you a small glass and a container (typically made of wood). (S)he will then put the glass into the container and start pouring sake into the glass. The server will fill the sake glass completely; in fact, (s)he will not stop until some sake overflows into the wooden container. This type of serving is a mark of generosity of the bar and the extra sake in the container is a bonus for you to enjoy (see **Figure 1**). You notice that it takes 5 seconds for the server to fill the sake glass, then the server continues to pour for an extra 1 second to make sure the glass is filled completely; the extra sake settles in the bottom of the wooden container.

Figure 1. Sake in a glass. http://everyonestea.blogspot.com/2012/03/cheerful-drink.html

a. (**25 points**) Since you know the dimensions of the glass and that of the container (see **Figure 2**) you wonder how much extra sake (*i.e.*, beyond filling the glass container) was served to you. Calculate this extra volume.

b. (**10 points**) You also wonder if you can calculate the height of the extra sake in the bottom container (corresponding to the extra volume of sake). While enjoying the lovely drink, you get to work and set up equations that will help you determine the level of sake in the bottom (wooden) container. In your calculations assume that the thickness of the glass is infinitesimally thin (*i.e.*, you can ignore the thickness of the glass container).

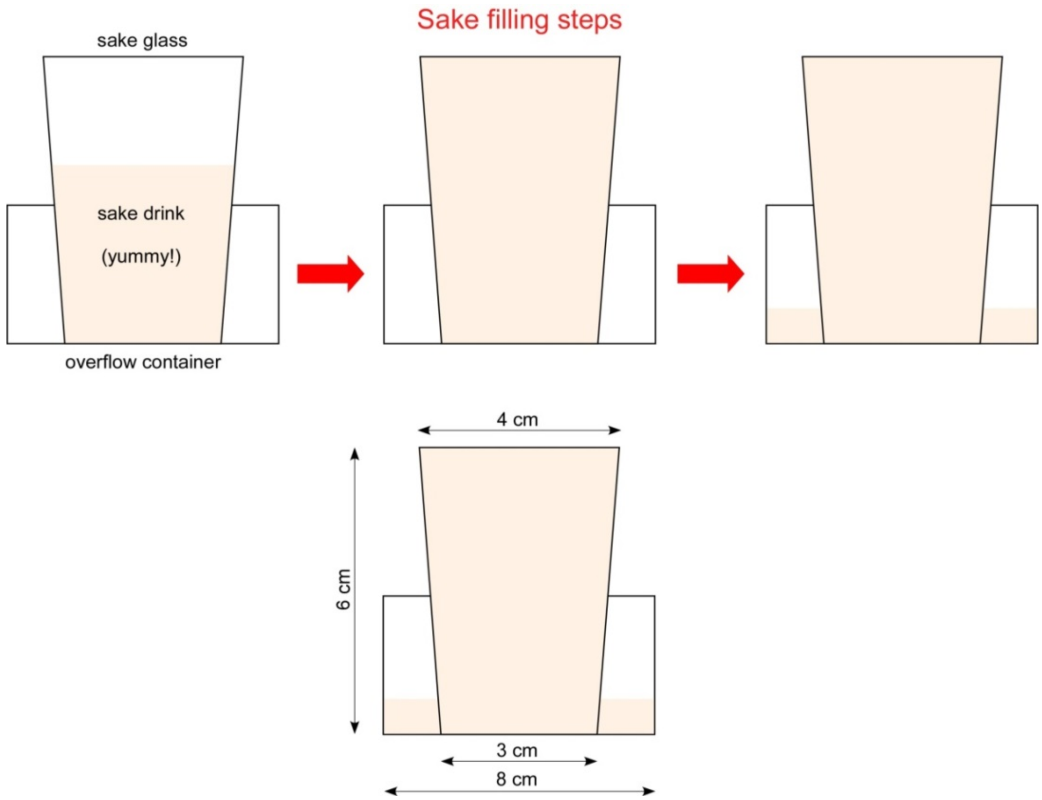

Figure 2. Description of sake glass filling process. You get the dimensions of the containers from this figure.

Problem 2 (**35 points total**)
The following chemical reactions take place in a liquid-phase batch reactor of constant volume V.

$A \xrightarrow{k_x} 3B$ (1st order reaction), k_x=0.1 s^{-1} (*i.e.*, $\frac{dc_A}{dt} = -k_x c_A$)

$B \xrightarrow{k_y} C$ (2nd order reaction), k_y=0.5 $s^{-1} \cdot M^{-1}$ (*i.e.*, $\frac{dc_C}{dt} = k_y (c_B)^2$)

The reactor is initially filled with a solution of A having a concentration of C_{A0}=0.1 mol/L. No B and C species are present at time=0 seconds.

a. (**21 points**) Using the order of chemical reaction we learned in the class, derive differential equations of dC_A, dC_B, and dC_C as a function of dt (e.g., dC_A/dt=, dC_B/dt=, and dC_C/dt =). Plug in known variables (e.g., k_x and k_y) in your differential equations. Provide the initial conditions for each differential equation.

b. (**10 points**) Solve the differential equation of dC_A/dt, and derive a final expression of C_A as a function t

c. (**4 points**) What is the expected concentration of C at times approaching infinity (that is, when all A gets converted into B and all B gets converted into C, what is the concentration of C)? [You should be able to get an answer to this without doing any calculations…].

Problem 3 (**30 points total**)

A continuous stirred tank reactor (CSTR) contains 10 L of A solution having a concentration of C_A=2 mol/L when time t=0 second (**Figure 3**). No other species inside the tank when the process starts (C_B(t=0)=0 and C_C(t=0)=0). At time t=0 seconds, the stank starts to be filled with two input streams. One stream contains 10 mol/L of A and is supplied at a volumetric flow rate of $\dot{V}_1$=0.15 L/s. The other stream contains 5 mol/L of B and is supplied at a volumetric flow rate of $\dot{V}_2$=0.2 L/s. The output stream leaves the CSTR at a volumetric flow rate of $\dot{V}_3$=0.35 L/s. Inside the tank, two chemical reactions occur when A meets B:

$$A+2B \rightarrow C \quad (k_1 = 0.1\ s^{-1},\ 1^{st} \text{ order reaction})$$

$$A \rightarrow 3B \quad (k_2 = 0.3\ M^{-1} * s^{-1},\ 2^{nd} \text{ order reaction})$$

where A and B produce new compound C; In the meantime, A will also decompose to generate B. For the main reaction, the rate of reaction r can be expressed as following:

$$r = k_1 C_A^{0.5} C_B^{0.5}$$

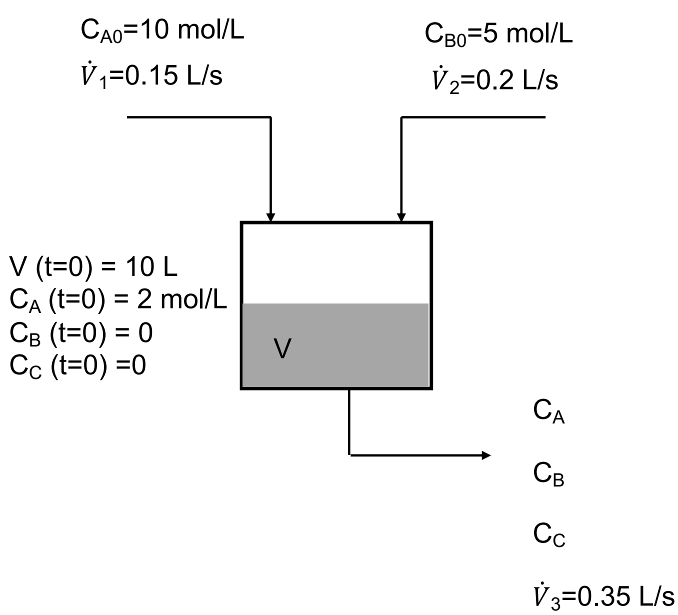

Figure 3. Schematic of the CSTR

Based on the above information, starting with the transient mole balance, derive three differential equations of dC_A, dC_B, and dC_C as a function of dt (e.g., dC_A/dt=, dC_B/dt=, dC_C/dt=). Plug in known variables, such as $\dot{V}_1$, $\dot{V}_2$, $\dot{V}_3$, k_1, k_2, V, C_{A0}, etc. Specify initial conditions for each differential equation. DO NOT SOLVE THE DIFFERENTIAL EQUATIONS.